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Master's End-of-Study Project (PFE)
Master in **[TODO: your master programme name]**

Learning Offensive Navigation Policies on Heterogeneous Attack Graphs

A GNN Approach to Sequential Privilege Escalation
in Active Directory

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Academic Year 2025 – 2026

Acknowledgements

First and foremost, I would like to express my deepest gratitude to my academic supervisor GHADI Abderahim, for his unwavering guidance, patience, and invaluable feedback throughout this academic year.

I am equally grateful to my professional supervisor, ZAABOUL Jawad, and the entire team at NearSecure. Thank you for providing an environment that fosters innovation, for trusting me to pursue the intersection of offensive security and Deep Learning, and for the technical mentorship that bridged the gap between academic theory and real-world application.

A massive thank you is owed to the broader cybersecurity and open-source communities. The development of the GNN-AD-Navigator would not have been possible without the tireless work of the developers who build and maintain the tools we rely on. Specifically, I would like to acknowledge:

SpecterOps, for creating and open-sourcing BloodHound, which provided the foundational graph topology and data schema necessary for my model's ingestion pipeline.

Oliver Lyak ly4k, for developing Certipy, an essential tool that allowed me to accurately map complex Active Directory Certificate Services ADCS attack paths into my heterogeneous graph.

Mayfly Orange Cyberdefense, for creating GOAD Game of Active Directory. The immense complexity of this lab environment served as the perfect training ground for the neural network, allowing it to learn and discover emergent tactical behaviors.

I also want to thank Hack The Box Academy for their student program, which granted me accessible access to the inlanefreight Active Directory environment. This provided the critical cross-environment validation required to prove the model's efficacy outside of its training data.

Finally, a personal thank you to my father and family for the ongoing support, advice and guidance during the long hours spent debugging and studying. To everyone who supported me through this Master's journey: thank you.

Abstract

Active Directory is a directory service so deeply embedded in corporate infrastructure that misconfigurations, no matter how small, can expose entire organisations to privilege escalation attacks. Tools like BloodHound are critical for both defenders and red teams: they map the environment as a graph and let operators query attack paths between any two objects. BloodHound’s built-in pathfinding, however, runs plain Dijkstra — it returns the shortest path, not necessarily the most operationally sound one, and it cannot learn from prior engagements.

This thesis asks whether a Graph Neural Network can learn a navigation policy from labelled attack traces and generalise it to entirely unseen enterprise environments. A Heterogeneous Graph Transformer is trained on a multi-domain lab environment and evaluated on a 4 000-node enterprise network it has never seen. The model recovers the expert-documented Domain Admin attack path without retraining, establishing cross-environment generalisation as a proof of concept. Identified architectural boundaries and a forward-looking hypothesis for a second-generation tool are discussed.

Keywords: Graph Neural Network, Heterogeneous Graph Transformer, Active Directory, Privilege Escalation, Attack Graph, Beam Search, Offensive Security, Cross-Environment Generalisation, Active Directory Certification Services.

Résumé

Active Directory est un service d'annuaire si profondément intégré dans l'infrastructure des entreprises que la moindre erreur de configuration peut exposer une organisation entière à des attaques de privilege escalation. Des outils comme BloodHound sont essentiels à la fois pour les défenseurs et les red teams : ils cartographient l'environnement sous forme de graphe et permettent aux opérateurs de rechercher des chemins d'attaque entre deux objets. Cependant, le moteur de recherche de BloodHound utilise le simple algorithme de Dijkstra renvoie le chemin le plus court, et non le plus pertinent d'un point de vue opérationnel.

Cette thèse cherche à savoir si un Graph Neural Network peut apprendre une politique de navigation à partir de traces d'attaques étiquetées, et la généraliser à des environnements d'entreprise totalement inconnus. Un Heterogeneous Graph Transformer est entraîné sur un environnement de laboratoire multi-domaines, puis évalué sur un réseau d'entreprise de 4 000 nœuds qu'il n'a jamais vu auparavant. Le modèle parvient à retrouver le chemin d'attaque Domain Admin documenté par des experts, et ce sans réentraînement, validant ainsi la généralisation cross-environment comme un véritable Proof of Concept. Enfin, les limites architecturales identifiées ainsi que des pistes d'évolution pour un outil de seconde génération sont discutées.

Mots-clés : Réseau de neurones sur graphe, Transformateur de graphe hétérogène, Active Directory, Escalade de privilèges, Graphe d'attaque, Recherche en faisceau, Sécurité offensive, Généralisation inter-environnements.

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List of Abbreviations

ADCS Active Directory Certificate Services. 2

DCSync Directory Replication Service synchronisation attack. 2

GOAD Game of Active Directory. 2

HGT Heterogeneous Graph Transformer. 2

PyG PyTorch Geometric. 2

RBCD Resource-Based Constrained Delegation. 2

SID Security Identifier. 2

General Introduction

Context

Active Directory is the identity backbone of major enterprises. While typically isolated within the inside network, once the perimeter defenses are breached (e.g., via web service exploitation, phishing, or a compromised usb) an adversary gains the ability to directly interact and query the Active Directory Domain Controller. Through protocols such as LDAP and SMB the entirety of the organisation's IT infrastructure, data and services is essentially mapped out for the attacker. A vulnerable AD poses extreme levels of risk making hardening the AD network environment an extreme priority.

Once inside, attackers frequently leverage credential-based and ACL-based attacks such as pass-the-hash, DCSync, Kerberoasting are among the most impactful attack vectors in modern breaches. Red teams and security auditor alike rely on tools like BloodHound to map the attack surface and identify the vulnerable nodes during an engagement. However the analysis remains a highly manual process, time consuming, heavily dependent on human expertise and none scalable in large enterprise environments.

Problem Statement

Bloodhound maps the entirety of the Active Directory to a graph database, each node represents an object, and the relationships between these objects is represented by edges. When a user runs a path finding query, BloodHound simply runs Dijkstra over the attack graph, it finds the shortest path, not necessarily the most realistic or discreet one.

The question this thesis asks : can a model learn a navigation policy and generalise that policy to an unseen enterprise environment?

Objectives

The objective of this Thesis is to provide a proof of concept for learned attack path discovery over heterogeneous Active Directory graphs.

To that end this work aims to:

- Build an end-to-end pipeline transforming raw BloodHound and Certipy JSON

to navigable [PyTorch Geometric \(PyG\)](#) [HeteroData](#) tensors.

- Train a GNN-based navigation policy and demonstrate cross-environment generalisation to unseen enterprise level environments.
- Identify and precisely describe the model’s limitations and architectural boundaries through systematic evaluation, establishing future work and improvements.

Main Contributions

- A complete, modular heterogeneous graph pipeline from raw BloodHound/Certipy JSON to navigable [PyG HeteroData](#) tensors, covering [Active Directory Certificate Services \(ADCS\)](#), [Resource-Based Constrained Delegation \(RBCD\)](#), [GPLink](#), and [Security Identifier \(SID\)](#) history edges.
- A [Heterogeneous Graph Transformer \(HGT\)](#)-based navigation policy trained on [Game of Active Directory \(GOAD\)](#) traces and evaluated on the entirely unseen [INLANEFREIGHT](#) enterprise environment, demonstrating generalisation from a 351-node lab to a 4 000-node network.
- A diagnostic evaluation of the learned policy’s architectural boundaries identifying how each design oversight [Directory Replication Service synchronisation attack \(DCSync\)](#) goal-state blindness, computer-node edge directionality, and training-distribution sparsity propagated into observable failure modes, and estimating the performance ceiling imitation learning could reach if these constraints were lifted.
- A forward-looking system hypothesis describing a second-generation tool where identified oversights are addressed: a goal-conditioned policy with [DCSync](#)-aware terminal learning, bidirectional computer-node traversal, and an expanded training corpus outlining what the approach is capable of reaching beyond the bounds of this proof of concept.

Report Outline

The remainder of this document is structured as follows. Chapter 1 establishes the theoretical foundation, reviewing the architecture of Active Directory, common privilege escalation techniques, the machine learning logic behind GNNs and the current state of graph-based attack path analysis. Chapter 2 details the methodology and system architecture, introducing the heterogeneous graph pipeline designed to transform raw BloodHound data into navigable tensors and present the core logic of the navigation policy the loss function and the beam search method. Chapter 3 presents the architecture and training paradigm of the Graph Neural Network (GNN) navigation policy. Chapter 4

provides an empirical evaluation of the learned policy across multiple environments, documents both successful generalisations and precise failure modes, and identifies the architectural oversights that bound the current approach. Chapter 5 concludes the report by reviewing what this proof of concept achieved and laying out its limitations honestly. It then builds a hypothesis where a second generation tool, one where the identified oversights are properly handled, could theoretically reach as well as pinpointing where future research might target.

Chapter 1

Background and Related Work

1.1 Active Directory: Architecture and Security Model

1.1.1 Core Objects and Structure

Active Directory (AD) is a directory service for Windows network environments. It is a distributed, hierarchical structure that allows for centralized management of an organization's resources, including users, computers, groups, network devices, file shares, group policies, servers and workstations, and trusts. AD provides authentication and authorization functions within a Windows domain environment.¹

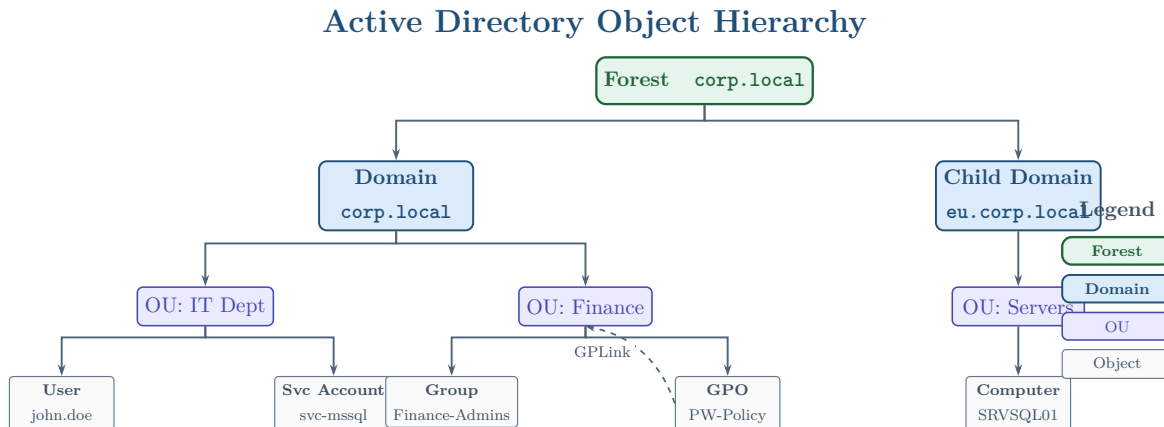


Figure 1.1: Active Directory object hierarchy.

A basic AD user account with no added privileges can be used to enumerate the majority of objects contained within AD, including but not limited to

- Domain Computers
- Domain Group Information
- Default Domain Policy
- Password Policy

- Domain Trusts
- Domain Users
- Organizational Units (OUs)
- Functional Domain Levels
- Group Policy Objects (GPOs)
- Access Control Lists (ACLs)

1.1.2 Authentication Protocols

[TODO: Write authentication section]

1.1.3 Trust Relationships

[TODO: Write trust relationships section]

1.2 Privilege Escalation in AD Environments

1.2.1 Offensive ACL Abuse

[TODO: Write ACL abuse section]

1.2.2 DCSync Attack

[TODO: Write DCSync section]

1.2.3 ADCS Attack Paths

[TODO: Write ADCS section]

1.2.4 SID History Abuse and RBCD

[TODO: Write SIDHistory and RBCD section]

1.3 Existing Tools and Their Limitations

1.3.1 BloodHound and SharpHound

[TODO: Write BloodHound/SharpHound section]

1.3.2 bloodhound-python

[TODO: Write bloodhound-python limitations]

1.3.3 Other Automated Attack Path Tools

[TODO: Write related tools survey]

1.4 Graph Neural Networks

1.4.1 Message Passing and Neighbourhood Aggregation

[TODO: Write GNN fundamentals]

1.4.2 Graph Convolutional Network (GCN)

[TODO: Write GCN section]

1.4.3 Heterogeneous Graph Transformer (HGT)

[TODO: Write HGT section]

1.4.4 PyTorch Geometric and HeteroData

[TODO: Write PyG section]

1.5 Summary and Positioning

While recent work has explored GNN-based modelling of attacker behaviour in AD graphs from a *defensive* perspective — approximating attacker decision-making to guide edge-blocking strategies **goel2025coevolutionary** — the complementary offensive direction remains largely unexplored. This thesis addresses that gap by learning an offensive navigation policy directly from labelled attack traces, evaluated on real enterprise environments rather than synthetic graphs. [TODO: Write summary and positioning]

Chapter 2

System Architecture and Design

2.1 System Overview

[TODO: Write system overview and include pipeline diagram]

2.2 Data Collection and Preprocessing

2.2.1 Input Sources

[TODO: Write input sources section]

2.2.2 `clean_bloodhound.py`: Filtering and Multi-Domain Accumulation

[TODO: Write `clean_bloodhound.py` section]

2.2.3 `merger.py`: Normalisation and Merging

[TODO: Write `merger.py` section]

2.2.4 `stitcher.py`: ADCS Injection and Domain Detection

[TODO: Write `stitcher.py` section]

2.3 Heterogeneous Graph Construction

2.3.1 Node and Edge Type Schema

[TODO: Write node/edge schema — include a table]

2.3.2 Direction Convention

[TODO: Write direction convention section]

2.3.3 Edge Extraction Details

[TODO: Write edge extraction details]

2.3.4 PyG HeteroData Encoding

[TODO: Write PyG encoding section]

2.4 Navigation Model

2.4.1 HGT Encoder

[TODO: Write HGT encoder section]

2.4.2 Policy Head

[TODO: Write policy head section]

2.4.3 Loss Function and Step Cost

[TODO: Write loss function section]

2.5 Beam Search Navigation

2.5.1 Algorithm

[TODO: Write beam search algorithm description — consider an Algorithm2e pseudocode block]

2.5.2 Terminal State: DCSync

[TODO: Write terminal state section]

2.5.3 Cross-Environment Inference

[TODO: Write cross-environment inference section]

2.6 Audit Module

[TODO: Write audit module section]

Chapter 3

Implementation

3.1 Technical Stack

[TODO: Write technical stack section — include a versions table]

3.2 Repository Structure

[TODO: Write repository structure section — include the directory tree as a listing]

3.3 Data Pipeline

3.3.1 Script Responsibilities and Data Flow

[TODO: Write pipeline data flow section — include input/output table]

3.3.2 SharpHound Version Normalisation

[TODO: Write SharpHound normalisation section]

3.4 Training

3.4.1 Training Data: GOAD

[TODO: Write GOAD training data section]

3.4.2 Kaggle Training Notebook

[TODO: Write Kaggle notebook section]

3.4.3 Model Checkpoints

[TODO: Write model checkpoints section]

3.5 Inference Pipeline

[TODO: Write inference pipeline section]

3.6 Graphical Interface

3.6.1 Design and Theme

[TODO: Write GUI design section]

3.6.2 Workflow Integration

[TODO: Write GUI workflow section]

3.6.3 Visualisation

[TODO: Write GUI visualisation section]

Chapter 4

Experiments and Results

4.1 Experimental Setup

4.1.1 Training Environment: GOAD

Game of Active Directory, a deliberately vulnerable Active Directory lab, created for penetration testers to practice common infrastructure attacks and privilege escalation techniques. GOAD consists of three domains — `sevenkingdoms.local`, `north.sevenkingdoms.local`, and `essos.local` — interconnected through domain trust relationships, mimicking a realistic multi-domain enterprise architecture. The lab exposes various sorts of misconfigurations commonly found in enterprise environments, from unpatched system vulnerabilities to unprotected services and network misconfigurations. The attack paths goad provides are to be presented in a learnable and readable training format for the model. Each attack path is deconstructed into transitions, `current_node`, `next_hop`, `credentials_used`, `total`, `path_weight`, `train_on_step`, `notes` and the `step_cost`.

4.1.2 Feature engineering

To a GNN model learning from a bloodhound scan, attack paths are of different importance, some are totally invisible while others are a gold mine for what the model needs to replicate. Paths that rely on unpatched vulnerabilities or outdated systems, for instance, leave no trace in the BloodHound graph and are marked with a `path_weight` of 0. Paths that are mostly visible with only two or less transitions that do not qualify have the latter marked through `train_on_step` set to zero. And at last, for an imitation learning model, given the choice between two valide transitions, the model choses based on training data, preferring the edge type it has seen most during training. To address this We are to more or less guide the model to gravitate towards the edge attack with less impact on the environment, the one with less noise and impact. To reach the desired effect each valid transition is scored. This score is reflected directly in the loss function through the `step_cost` feature.

In summary, the feature engineering phase introduced three mechanisms: filters

to eliminate CVE-dependent transitions invisible to the graph collector; path weights reflecting a composite visibility score that prioritises transitions where the graph structure itself is the primary escalation vector; and a `step_cost` feature guiding the model toward stealthier, less impactful edges when a choice presents itself. This constitutes a simplified reward shaping approach, consistent with standard practices in the imitation learning literature.

4.1.3 Test Environment: INLANEFREIGHT

INLANEFREIGHT is an enterprise Active Directory environment provided by HTB Academy as part of their penetration testing learning paths. Unlike GOAD, INLANEFREIGHT was never used during training making the results obtained on it a zero-shot generalisation. The environment spans multiple domains connected through cross-domain trust relationships, with a total of approximately 4 000 nodes, significantly larger and more complex than the training environment.

Data collection was performed using `bloodhound-python` for both testing and training environments, which produced the BloodHound JSON files later processed by the pipeline. ADCS data was collected separately using Certipy and injected into the graph at the stitching stage. It is worth noting that `bloodhound-python` has known collection limitations compared to SharpHound C#, particularly around cross-forest foreign group membership, a constraint that directly affected one of the queries evaluated in Section 4.5.

4.1.4 Diagnostic Environment: GOAD

While INLANEFREIGHT provides a realistic enterprise environment offers only for one targeted documented path to compromise, same goes for the child domain LOGISTICS.INLANEFREIGHT.LOCAL and the same forest trusted domain FREIGHT-LOGISTICS.LOCAL each of them with a specific cross domain attack path. The domain works well for generalisation validation but it is insufficient for dissecting model behaviour in depth.

GOAD, despite being the training environment, serves a second purpose here as a diagnostic environment. Its multi-domain structure and richer set of documented attack paths make it better suited for probing the model’s limitations, identifying where it fails and why. Queries run against GOAD in this context are not used to measure generalisation (the model has seen this environment) but rather to isolate specific architectural weaknesses under controlled conditions.

4.1.5 Evaluation and Limitation Identification Logic

The evaluation methodology follows a continuous prompting approach: the model is repeatedly queried across different start and target node pairs, and its output is compared against BloodHound’s built-in pathfinding on the same graph. The choices the model makes at each step, combined with knowledge of the pipeline’s collected data and the `bloodhound-python` injector’s known limitations, allow for precise isolation of each failure mode and a confident proposal of remediation actions for future work.

Queries are not filtered to show only successes. Failed queries, degenerate paths, and cases where the model’s output diverges significantly from BloodHound’s are reported alongside valid results. This approach avoids a cherry-picking reading of the results: the model is evaluated at its coverage boundary, and failures are treated as diagnostic data.

4.2 Result 1: `wley` → Domain Admin on INLANEFREIGHT

Query and Context

To establish a baseline for the model’s navigational capabilities, the initial query targets a well-documented exploit chain within the INLANEFREIGHT enterprise environment. The objective is to navigate from the unprivileged user `wley` to achieve Domain Admin privileges. This scenario serves as a critical competency test: evaluating whether the model can apply sequential privilege escalation mechanics learned in a small, 351-node laboratory (GOAD) to a structurally complex, unseen 4 000-node enterprise graph.

Recovered Path

Operating purely on its learned policy with an `HGT` confidence score of 0.5620, the tool correctly provides the documented path. Navigates from the unprivileged user `wley` to reach the `domain admins` group. The tool given a new unseen Active Directory environment, navigates and recommends the same documented path, traversing edges the same way it had learned during training on this new architecture.

1. **Password Reset:** `wley` executes a `ForceChangePassword` attack on the user `damundsen`.
2. **Targeted Write:** `damundsen` abuses `GenericWrite` privileges to compromise the `help desk level 1` group.
3. **Group Inheritance:** This access provides an implicit `MemberOf` transition into the `information technology` group.

4. **Full Control to DCSync:** The information technology group exercises `GenericAll` over the user `adunn`.

Crucially, the pipeline’s audit module halts the beam search precisely at `adunn`. It verifies that this user natively holds the required rights (`GetChanges` and `GetChangesAll`) to execute a DCSync attack on the target domain, terminating the search with confirmed operational success.

```

1
2 Path 1:
3 Score: 0.5620 Steps: 4
4 [1] wley@inlanefreight.local
5     -> damundsen@inlanefreight.local
6     edge    : forcechangepassword
7     attack  : Password reset
8     action  : Force-change target's password
9 [2] damundsen@inlanefreight.local
10    -> help desk level 1@inlanefreight.local
11    edge    : genericwrite
12    attack  : ACL abuse - targeted write
13    action  : Modify SPN, msDS-
14              AllowedToActOnBehalfOfOtherIdentity
15 [3] help desk level 1@inlanefreight.local
16    -> information technology@inlanefreight.local
17    edge    : memberof
18    attack  : Group inheritance
19    action  : Inherit rights via group membership
20 [4] information technology@inlanefreight.local
21    -> adunn@inlanefreight.local
22    edge    : genericall
23    attack  : ACL abuse - full control
24    action  : Reset password, add SPN, modify object
25
26 AUDIT:
    Path terminates at a node with DCSync on the target domain
    'inlanefreight.local'. This is operational success.
```

Listing 4.1: model output for escalation path on INLANEFREIGHT

4.3 Result 2: Cross-Domain Trust Traversal


```
1 Start : htb-student_adm@logistics.inlanefreight.local (users)
2 Target : domain admins@inlanefreight.local (groups)
3
4
5 Path 1:
6   Score: 0.2349 Steps: 6
7   [1] htb-student_adm@logistics.inlanefreight.local
8       -> domain admins@logistics.inlanefreight.local
9       edge : memberof
10      attack : Group inheritance
11      action : Inherit rights via group membership
12   [2] domain admins@logistics.inlanefreight.local
13       -> logistics.inlanefreight.local
14       edge : writedacl
15      attack : ACL abuse - DACL modification
16      action : Grant yourself GenericAll on target
17   [3] logistics.inlanefreight.local
18       -> inlanefreight.local
19       edge : sameforesttrust
20      attack : Same-forest trust traversal
21      action : Cross domain via SID History
22   [4] inlanefreight.local
23       -> freightlogistics.local
24       edge : crossforesttrust
25      attack : Cross-forest trust traversal
26      action : Inter-forest pivot
27   [5] freightlogistics.local
28       -> domain controllers@freightlogistics.local
29       edge : contains
30      attack : Container hierarchy
31      action : Object lives inside OU/container
32   [6] domain controllers@freightlogistics.local
33       -> academy-ea-dc03.freightlogistics.local
34       edge : contains
35      attack : Container hierarchy
36      action : Object lives inside OU/container
37
38 AUDIT:
39 X Path terminates at 'academy-ea-dc03.freightlogistics.
   local' which has no DCSync rights
```

```

40     and is not the target. The model could not find a working
        attack chain.
41     ! Final node is in domain 'academy-ea-dc03.freightlogistics
        .local' but target is
42     in 'inlanefreight.local'. Cross-domain attack may need
        additional steps.

```

Listing 4.2: model output for cross-domain trust traversal on INLANEFREIGHT

The model navigated the cross domain trust edge, recommending the correct documented path. The processed graph portrays correctly the trust between domains. The model identified the edge type seen during training and traverses it. The rest of the path is noise a direct result of an architectural limitation. The specified target node `admins@inlanefreight.local` does not have a direct incoming path. When all the nodes are outgoing the model panics and produces noise. The audit script catches the errors and alerts the user.

4.3.1 Cross-Forest Foreign MemberOf

```

1
2 Start  : administrator@inlanefreight.local (users)
3 Target : administrators@freightlogistics.local (groups)
4
5 Path 1:
6     Score: 0.0918  Steps: 1
7     [1] administrator@inlanefreight.local
8         -> administrators@freightlogistics.local
9         edge      : memberof
10        attack    : Group inheritance
11        action    : Inherit rights via group membership
12
13 AUDIT:
14     Path reaches the requested target directly.

```

Listing 4.3: Model output on unseen node

One common misconfiguration in Active Directory forests, when the team with admin rights responsible for Active directory and network configurations, gives the admins of two domains membership of the same group. The admin account of one domain now holds admin rights over the other domain, allowing for easier interventions for the admin to adjust any configuration within the other domain withing the same forest. The pipeline manages to extract the `cross-forest Foreign MemberOf` edge

presenting it to the model. Once seen during training traversing the edge, the tool outputs the correct ACL abuse path to the user.

4.4 Result 3: Constrained Delegation

```

1 Start   : castelblack.north.sevenkingdoms.local (computers)
2 Target  : kingslanding.sevenkingdoms.local (computers)
3
4 Path 1:
5   Score: 1.0000 Steps: 2
6   [1] castelblack.north.sevenkingdoms.local
7       -> winterfell.north.sevenkingdoms.local
8       edge   : allowedtodelegate
9       attack : Constrained Delegation
10      action : S4U2Self + S4U2Proxy impersonation
11   [2] winterfell.north.sevenkingdoms.local
12       -> enterprise.domain.controllers@sevenkingdoms.local
13       edge   : memberof
14       attack : Group inheritance
15       action : Inherit rights via group membership
16
17 AUDIT:
18   Path terminates at a node with DCSync on the target domain
19   'kingslanding.sevenkingdoms.local'. This is operational
20   success.

```

Listing 4.4: Model output on unseen node

Key insight to foreground, the `AllowedToDelegate` edge was missed during the graph construction during training. The `AllowedToDelegate` is an edge type not directly present in the bloodhound scan, it needed further processing deducing the relationship between the two edges before establishing its existence. Hence during training, the model did not see this edge type. When introduced for this query as the only navigable node, the model suggested traversing it and gave a score of 1 (the only probable path).

4.5 Result 4: Documented Limitations

castelblack → *braavos.essos.local*

This query produces an interesting divergence between the model’s output and Blood-Hound’s pathfinding. The model navigates: *castelblack* → *winterfell* (`AllowedToDelegate`)

→ enterprise domain controllers@sevenkingdoms (MemberOf) → sevenkingdoms.local (GetChanges, partial DCSync) → essos.local (CrossForestTrust) → laps@essos (Contains) → braavos.essos.local (Contains).

BloodHound identifies a structurally different route passing through `CoerceToTGT` on `north.sevenkingdoms.local`, followed by a `SameForestTrust` pivot and an ACL chain through `tyron.lannister` (`GenericWrite`) and `ReadLAPSPassword` toward the target.

The divergence is explained by two limitations working together. First, `CoerceToTGT` is a protocol-level coercion primitive invisible to `bloodhound-python` and absent from the training data entirely — the model has no learned signal for this edge type and cannot reproduce BloodHound’s route. Second, the two `Contains` hops at the end of the model’s path are structural noise: the model reaches `essos.local` via a valid `CrossForestTrust` pivot but continues walking into container hierarchy rather than stopping, reflecting the terminal state blindness discussed in Section 5.3.

Neither path is strictly superior. The model’s route relies on a partial DCSync and a cross-forest trust traversal — a valid operational chain. BloodHound’s route relies on `CoerceToTGT`, which requires active protocol-level interaction beyond what static graph analysis can represent. This result illustrates both the boundaries of graph-only navigation and the gap left by `bloodhound-python`’s collection limitations.

[TODO: Write Result 4]

4.6 Discussion

4.6.1 Model Coverage and Training-Set Ceiling

[TODO: Write coverage discussion]

4.6.2 Expressiveness vs. Transferability: HGT vs. GCN

[TODO: Write HGT vs. GCN discussion]

4.6.3 Pipeline Completeness as a First-Class Contribution

[TODO: Write pipeline completeness discussion]

Chapter 5

Limitations and Future Work

5.1 Data Collection Incompleteness

[TODO: Write data collection limitations section]

5.2 Expanding the Training Knowledge Base

[TODO: Write training knowledge base expansion section]

5.3 Terminal State Learning via Offline Reinforcement Learning

[TODO: Write offline RL future direction section]

5.4 Additional Future Directions

5.4.1 Calibrated Confidence Scoring

[TODO: Write calibrated confidence section]

5.4.2 SharpHound Edge Vocabulary Versioning

[TODO: Write edge vocabulary versioning section]

5.4.3 Derived Attack Primitives

[TODO: Write derived primitives section]

General Conclusion

[TODO: Write general conclusion (1 page)]

Appendix A

Sanitised BloodHound JSON Example

[TODO: Insert sanitised JSON excerpt as a `lstlisting`]

Appendix B

Heterogeneous Graph Schema

[TODO: Insert full graph schema table]

Appendix C

Key Code Excerpts

Effective Weight Loss Function

```
1 # eff_weight combines path importance with step discretion cost
2 eff_weight = path_weight * (1 - step_cost)
```

Listing C.1: Effective weight computation in `prepare_training_examples.py`

[TODO: Add beam search loop and `has_dcsync_on` excerpts here]

Appendix D

GUI Screenshots

[TODO: Insert GUI screenshots]